

Research:

[Single-cell analysis reveals insights into epithelial abnormalities in ovarian endometriosis: Cell Reports](#)

- **Study Overview & Approach:** The authors performed single-cell RNA sequencing (STRT-seq) on 7,030 individual cells across 34 women (23 with ovarian endometriosis, 11 healthy controls) to construct a high-precision single-cell atlas across both proliferative and secretory menstrual phases.
PDF+ 1
- **Increased Ciliated Cell Population:** Ciliated cells are present in higher proportions in both the eutopic (uterine cavity) and ectopic (ovarian lesion) endometrium of endometriosis patients compared to healthy controls.
PDF
- **Survival Advantage via Decreased SULT1E1:** Ciliated cells in patient tissues exhibit reduced expression of *SULT1E1* (estrogen sulfotransferase), leading to higher active estradiol levels, upregulation of anti-apoptotic genes (e.g., *BCL2*), and reduced apoptosis.
PDF
- **Apoptosis Resistance via NNMT Pathway:** Epithelial cells in ectopic lesions overexpress nicotinamide N-methyltransferase (*NNMT*), which stabilizes SIRT1 deacetylase, promotes deacetylation and nuclear exclusion of FOXO1, and downregulates the pro-apoptotic gene *BIM*, thereby allowing lesion cells to resist cell death.
PDF
- **Epithelial-Immune Regulation & Chronic Inflammation:** Ectopic epithelial cells display immune-like properties, elevated HLA class II complex expression, and chemokine secretion (*CXCL10*, *IL-15*). These cells directly recruit and activate CD4+ T cells while suppressing regulatory T cells (Treg), driving chronic inflammation in ectopic sites.
PDF+ 1
- **Stromal Cell Alterations:** Eutopic endometrial stromal cells upregulate pathways related to insulin response and MAPK signaling, providing further pro-survival and proliferative signals before tissue translocates to ectopic sites.
PDF

[E-MTAB-15117 < ArrayExpress < BioStudies < EMBL-EBI](#)

- This dataset originates from a study titled "**Global RNA expression analysis of patient samples identified potential diagnostic biomarkers specific for peritoneal, ovarian and deep endometriosis**". Endometriosis affects roughly 10% of women of reproductive age worldwide, but current diagnosis relies on invasive procedures (e.g., laparoscopy) and is often delayed by up to 10 years. This study uses RNA sequencing to identify

less-invasive, highly specific diagnostic biomarkers across distinct endometriosis subtypes.

- **Organism:** *Homo sapiens* (Human)
 - **Study Type:** RNA-seq of coding RNA (Sequencing assay)
 - **Sample Count:** 48 assays/samples total
- **Subtype Differentiation:** High-throughput transcriptomic profiling identified distinct gene expression signatures across three main endometriosis subtypes:
 - Superficial peritoneal endometriosis
 - Ovarian endometriosis
 - Deep infiltrating endometriosis
- **Immune Environment:** The dataset highlights significant variation in immune cell composition and inflammatory responses within lesion tissue compared to healthy control endometrial tissue.
- **Biomarker Discovery:** Serves as a diagnostic reference set to advance non-invasive or minimally invasive biomarker screening strategies.
- Investigation Format:
<https://ftp.ebi.ac.uk/biostudies/fire/E-MTAB-/117/E-MTAB-15117/Files/E-MTAB-15117.idf.txt?key=5edbe92e-92e6-47bf-9d06-3984660d223a>
- Sample and Data Format:
<https://ftp.ebi.ac.uk/biostudies/fire/E-MTAB-/117/E-MTAB-15117/Files/E-MTAB-15117.sdrf.txt?key=5edbe92e-92e6-47bf-9d06-3984660d223a>